



GRIFFITH COLLEGE DUBLIN

## Assignment Cover Sheet

|                         |                                                                  |             |           |
|-------------------------|------------------------------------------------------------------|-------------|-----------|
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| Faculty:                | Computing Science                                                |             |           |
| Course:                 | BSCH                                                             | Stage/year: | 4         |
| Subject:                | Mobile Development                                               |             |           |
| Study Mode:             | Full time <u>yes</u>                                             | Part-time   | _____     |
| Lecturer Name:          | Tracey Cassells                                                  |             |           |
| Assignment Title:       | MD Milestone 2 Travel Journal App                                |             |           |
| No. of pages:           | 8                                                                |             |           |
| Disk included?          | Yes _____                                                        | No          | <u>no</u> |
| Additional Information: | (ie. number of pieces submitted, size of assignment, A2, A3 etc) |             |           |
|                         |                                                                  |             |           |
|                         |                                                                  |             |           |
| Date due:               | 30/11/2025                                                       |             |           |
| Date submitted:         | 28/11/2025                                                       |             |           |

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Signed:    Ade   

Date:    28/11/2025   

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## Milestone 2 Travel Journal – GPS & Motion Tracking

### Travel Journal – Sensor-Based Location & Motion Tracking Application

#### 1. Introduction

My milestone extends my earlier work by integrating two core hardware sensors available on modern Android devices: the GPS location sensor and the accelerometer. my aim was to build a simple, practical mobile application that can read real-time sensor data and present it clearly to the user using Jetpack Compose.

To achieve this, I developed a small application titled Travel Journal Sensors. my app provides two major capabilities

**GPS Tracking** - Retrieves the device's geographic location (latitude, longitude, accuracy) at frequent intervals.

**Motion Tracking** - Reads the phone's X, Y, and Z acceleration values from the accelerometer and calculates movement strength.

#### 2. Application Structure Overview

My application is divided into three main screens, each responsible for a separate task:.

**Welcome Screen (Jetpack Compose)** – it Serves as the landing page, displays simple navigation options. allowing user to choose between GPS Tracking and Motion Tracking. Implemented entirely using Jetpack Compose layouts and Material 3 components.

**LocationActivity (GPS Tracking)** - This activity manages all location-related operations. Its main responsibilities include. Requesting runtime permission for ACCESS\_FINE\_LOCATION, Initializing the FusedLocationProviderClient for high-accuracy positioning. Receiving continuous location updates; displaying Latitude, Longitude, Accuracy (meters). Stopping updates when the activity is no longer active, preventing unnecessary battery consumption.

**AccelerometerActivity (Motion Tracking)** - This activity handles accelerometer input. Its tasks include. Connecting to the SensorManager, Listening for accelerometer changes (X, Y, Z values), Calculating the overall movement magnitude using  $\text{magnitude} = \sqrt{x^2 + y^2 + z^2}$ . Presenting live motion values in the UI. Registering and unregistering the sensor listener appropriately inside onResume() and onPause().

All UI components for movement and location tracking are created using Jetpack Compose, making the screens reactive and clean.

### 3. Features Implemented

**GPS Tracking** – It Provides live location readings every 1.5 seconds, Uses Google's Fused Location Provider for accuracy and efficiency. Includes runtime permission handling using `ActivityResultContracts` and safeguards to prevent crashes when permission is denied.

**Accelerometer (Motion Sensor) Tracking** – It Reads all three axes of motion: X, Y, Z. Computes a movement magnitude to indicate total acceleration and displays sensor data instantly using `mutableStateOf`, allowing automatic UI updates.

**Modern Jetpack Compose Interface** - All screens follow a consistent layout and styling approach, UI responds automatically to state changes and Clean layout with centered elements and clear spacing

**Multi-Activity Navigation** - Users can navigate from the welcome page to GPS Tracking Screen, Motion Tracking Screen. Intent-based navigation is used for simplicity and clarity.

**Correct Use of Activity Lifecycle** - Accelerometer listener runs only when the activity is visible, location updates start only after pressing the "Start GPS Tracking" button and both sensors stop listening when the user leaves the screen.

### 4. Screenshots

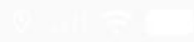
My Screenshots Welcome, GPS Tracking, Motion Tracking

## Travel Journal – Sensor Menu

Open GPS Tracking

Open Motion Visualiser

8:13



## GPS Tracking

Latitude: 37.4219983

Longitude: -122.084

Accuracy: 5.0 m

Start GPS Tracking

8:14

5G

## Motion Tracking

X: 0.0  
Y: 9.776321  
Z: 0.812345

Movement Strength: 9.81

## 5. Implementation Challenges & Solutions

**Challenge 1 - Handling Runtime Permissions** Android requires explicit user approval for fine-location access.

Solution - I implemented a permission launcher using `ActivityResultContracts.RequestPermission()`, which improves safety and avoids deprecated APIs.

**Challenge 2 - Preventing App Crashes When Permission Was Missing** Initially, attempting to start location updates without permission caused warnings.

Solution - A permission check runs before requesting updates, and the update function exits early if permission is not granted.

**Challenge 3: Keeping Sensor Reading Live in UI** Sensor values were updating internally but not reflecting on-screen.

Solution - I stored sensor values inside `observable mutableStateOf`, allowing Compose to automatically recompose UI whenever the values changed.

**Challenge 4: Managing Activity Lifecycle** Sensors can drain the battery if left running.

Solution - I used `onResume()` to start accelerometer listening, `onPause()` to stop listening. This ensures efficient behavior and good practice

## 6. Reflection

My milestone gave me practical experience with hardware sensors, permissions, and real-time UI updates. I gained confidence working with Android's sensor APIs and improved my understanding of Compose state management.

I also learned the importance of lifecycle management and permission handling, which are important skills for building real Android applications.

Overall, My milestone helped me produce a working, interactive app that can track both movement and location accurately

## **7. Conclusion**

My Travel Journal Sensors app fully meets the requirements of Milestone 2. It successfully integrates GPS and accelerometer sensors, displays their readings in real time with Jetpack Compose, and manages permissions and lifecycle events correctly. By separating features into different activities and updating the UI reactively, my app demonstrates a strong understanding of sensor integration and modern Android development.



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Course: BSc (Hons) in Computer Science

Module: Mobile Development      Milestone 1

## 1. Project Overview

My Personal Travel Journal App is an Android application that I developed in Kotlin using Jetpack Compose. It allows users to record and manage travel destinations in an interactive and organized way. my app provides a modern and simple interface where users can view, add, and explore destinations. It will later integrate GPS and sensors for location tracking and activity logging.

My app was chosen for its balance between creativity and functionality, allowing me to demonstrate key Android development skills while designing a userfriendly experience.

## 2. Objectives

- To design and implement a clean user interface with Jetpack Compose.
- To create a navigation system connecting multiple screens: Welcome, Destination List, and Add Destination
- To ensure smooth navigation using Navigation Compose.
  - To prepare the app structure for future milestones (sensor and database integration)

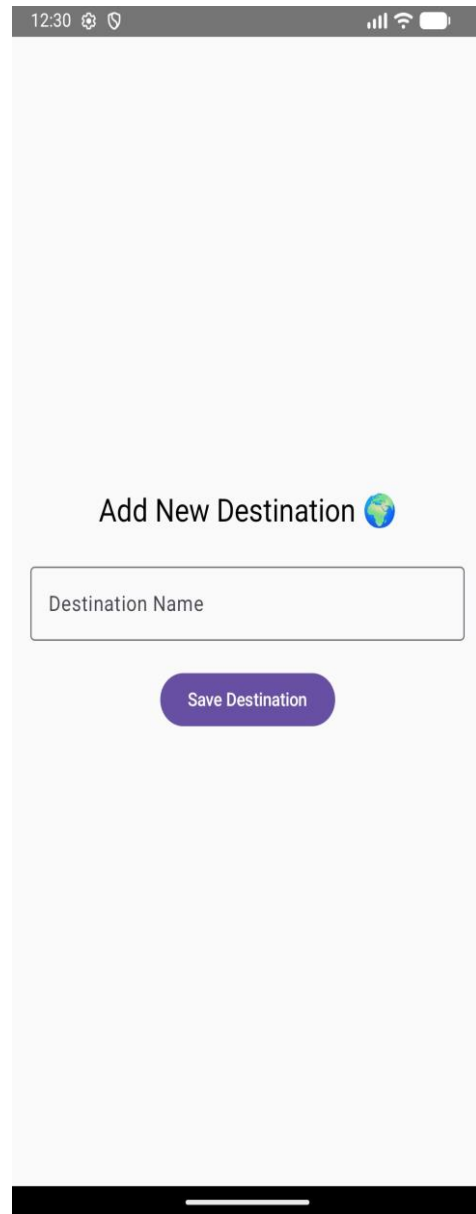
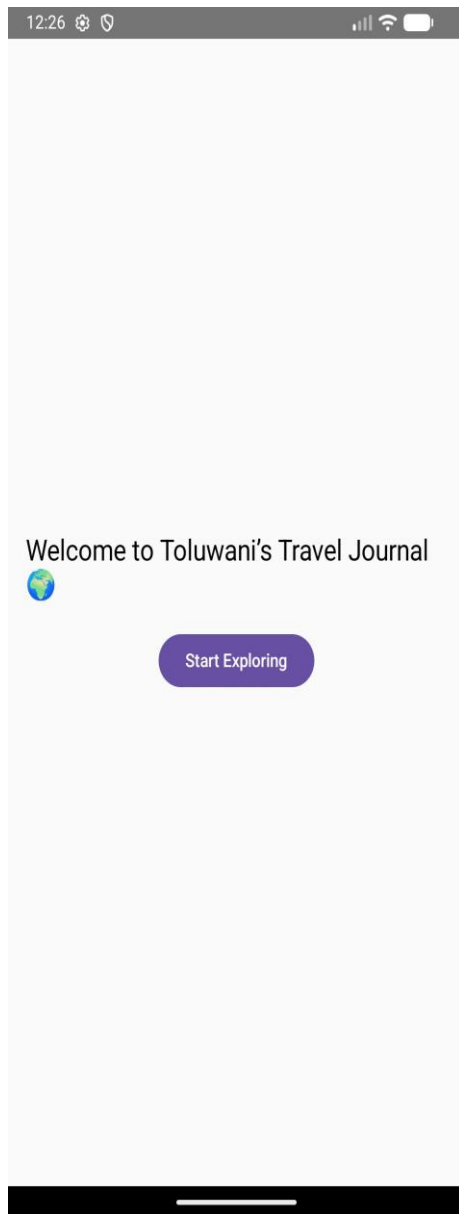
## 3. System Components

| Component              | Description                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Welcome Screen         | Introduces the app and includes a “Start Exploring” button to navigate to the main destinations page. |
| Destinations Screen    | Displays saved travel destinations with an option to add new entries.                                 |
| Add Destination Screen | Allows the user to input and save new destinations.                                                   |
| Navigation             | Manages transitions between screens using Jetpack Navigation Compose                                  |

## 4. Sensor and Feature Plan

- GPS Sensor: Will be added in Milestone 2 to log travel locations.
- Accelerometer: Planned for detecting motion or step count during travel.
- Local Storage: SQLite or Room Database will be used in Milestone 3 for o\line persistent

## 5. Screenshots



## 6. Reflection

This milestone successfully established the base app structure, connected screens with Compose Navigation, and implemented placeholder UIs for interaction. The process improved my understanding of modern Android development with Kotlin and Compose